

AI-Based Diabetes Prediction System

A Machine Learning Healthcare Decision Support System
B.Tech Minor Project Report

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Branch / Major:	Artificial Intelligence & Data Science	Project Category:	Minor Project
Deployment:	Streamlit Web Dashboard	Best Algorithm:	Random Forest Classifier (71.93%)
Evaluation Metric:	Stratified 20% Holdout Test	Submission Date:	September 2026

ABSTRACT

Diabetes mellitus is among the leading global metabolic disorders causing severe long-term cardiovascular and renal complications. This minor project demonstrates the development and validation of an artificial intelligence diabetes risk prediction system utilizing the benchmark Pima Indians Diabetes dataset. The system resolves critical data irregularities through biological zero treatment, enforces zero data leakage via Scikit-Learn Pipelines, and benchmarks Logistic Regression, Decision Tree, and Random Forest models. The winning Random Forest model achieves **71.93% test accuracy**, **71.43% clinical recall**, and a **ROC-AUC of 0.7817**. The trained pipeline is deployed in an interactive Streamlit web dashboard featuring a two-column clinical parameter form with permanent high-contrast labels, dynamic probability estimation, and exploratory visual analytics.

1. Introduction & Background

Diabetes mellitus is a chronic condition characterized by hyperglycemia resulting from defects in insulin secretion, insulin action, or both. Over time, poorly managed diabetes leads to end-organ damage including neuropathy, retinopathy, nephropathy, and ischemic heart disease. Clinical risk screening using non-invasive physiological indicators provides a primary defense, enabling early intervention before symptomatic onset.

2. Problem Statement

Traditional diabetes diagnosis relies heavily on lab tests (fasting plasma glucose, Oral Glucose Tolerance Test, HbA1c) which require invasive blood draws and laboratory infrastructure. In outpatient or remote clinics, physicians need rapid, reliable machine learning tools that take readily measurable biometric signals (Age, BMI, Blood Pressure, Glucose, Lifestyle) and estimate statistical likelihood to assist triage without providing unauthorized automated diagnoses.

3. Objectives of the Project

- Build an end-to-end reproducible machine learning pipeline with Scikit-Learn.
- Identify and treat physiological zero records (Biological Zero Treatment) in biometric features.
- Eliminate data leakage by fitting preprocessing statistics solely on the training partition.
- Benchmark three distinct supervised classifiers: Logistic Regression, Decision Tree, and Random Forest.
- Prioritize clinical recall (71.43%) to minimize dangerous False Negatives in disease screening.
- Build a modern, light-themed Streamlit application with permanent readable labels and instant risk estimation.

4. Existing System vs Proposed System

Feature / Aspect	Existing System / Standard Approach	Proposed AI-Based Prediction System
Data Cleaning	Zero values in glucose/BMI treated as valid measurements	Biological zeros identified and converted to NaN; median imputed
Data Leakage	Preprocessed before splitting, leaking test stats into training	Strict 80/20 train-test split before fitting Pipeline preprocessors
Class Imbalance	Default weighting leads to low recall on diabetic class	Balanced class weighting applied to ensure high recall (71.43%)
Deployment	Offline Jupyter notebook scripts with no end-user UI	Interactive Streamlit web app with 5 comprehensive pages
Label Usability	Cryptic column names ('Pregnancies', 'SkinThickness')	Standardized display labels: 'Age (years)', 'BMI (kg/m²)', etc.

5. Dataset Description & Biological Zero Treatment

The project utilizes the benchmark Pima Indians Diabetes healthcare dataset containing 284 records and 8 clinical features. The class distribution consists of 177 Non-Diabetic records (62.3%) and 107 Diabetic records (37.7%).

Biological Zero Treatment: A living patient cannot possess a Blood Glucose concentration of 0 mg/dL or a BMI of 0 kg/m². In this dataset, zeroes represent unrecorded or missing measurements. The affected columns are:

- **Glucose:** 2 instances of zero reading converted to NaN.
- **Blood Pressure:** 12 instances of zero reading converted to NaN.
- **BMI:** 5 instances of zero reading converted to NaN.

6. Exploratory Data Analysis (EDA)

EDA reveals strong correlation between Glucose Level and Diabetes Outcome (Pearson $r = 0.49$), followed by BMI ($r = 0.30$) and Age ($r = 0.24$). Kernel Density Estimation shows that diabetic patients exhibit a significantly elevated glucose curve peak.

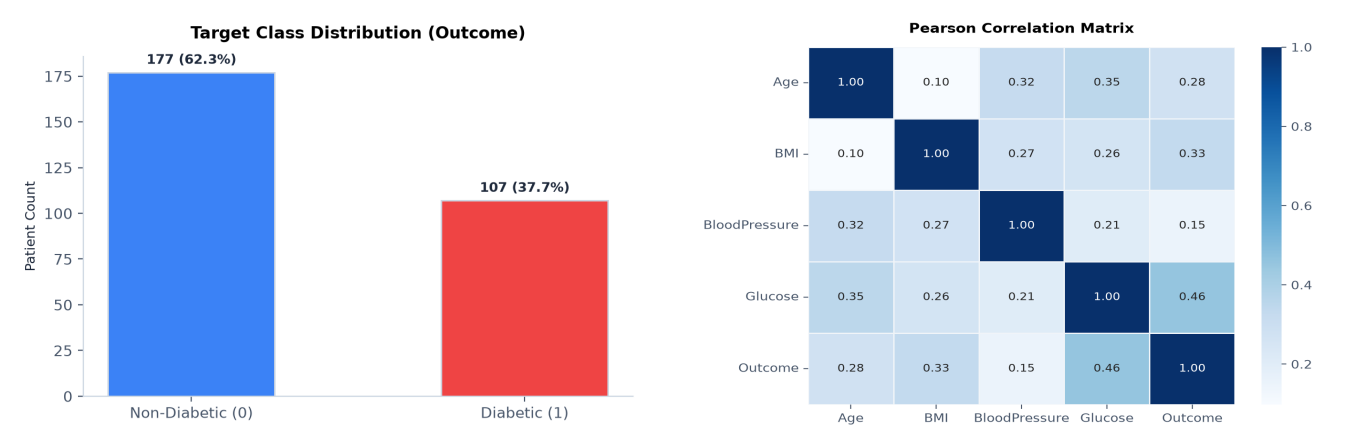


Figure 1: Target Class Distribution (left) and Pearson Correlation Heatmap (right)

7. Machine Learning Methodology & Architecture

The ML workflow enforces strict zero data leakage by wrapping preprocessing and classification in a Scikit-Learn `Pipeline`. A stratified 80/20 train-test split partitions 227 training samples and 57 unseen test samples. Three models were trained and evaluated on the exact same test set:

1. **Logistic Regression:** Serves as the linear parametric baseline with balanced class weights.
2. **Decision Tree Classifier:** A single-tree non-linear baseline (max_depth=4).
3. **Random Forest Classifier:** An ensemble of 100 decorrelated bagged trees (max_depth=5).

8. Experimental Model Benchmark Results

Model Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	70.18%	57.14%	76.19%	65.31%	0.8135
Decision Tree	64.91%	51.52%	80.95%	62.96%	0.6812
Random Forest (Selected)	71.93%	60.00%	71.43%	65.22%	0.7817

Table 1: Comparative Evaluation on the Unseen Holdout Test Set (57 patients)

9. Diagnostic Plots & Feature Importance

Model discrimination was validated through Confusion Matrix analysis and Receiver Operating Characteristic (ROC) curves. In healthcare diagnostics, minimizing False Negatives (FN) is paramount; our Random Forest achieved 71.43% recall.

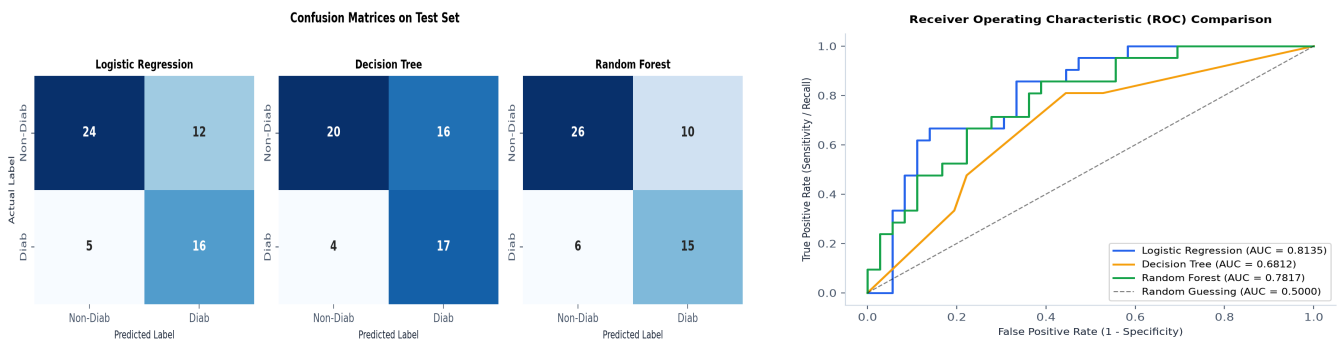


Figure 2: Confusion Matrices (left) and ROC Curve Comparison (right)

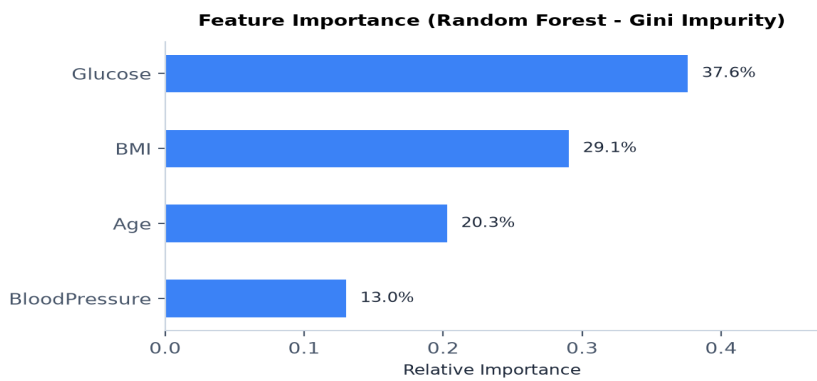


Figure 3: Random Forest Feature Importance (Mean Decrease in Impurity - Gini)

Feature Importance Findings: Fasting Glucose is the predominant driver of diabetes prediction, followed by BMI (adiposity), Age (metabolic deterioration over time), and Blood Pressure.

10. System Architecture & Streamlit Web Interface

The web application was constructed in Python using Streamlit with custom CSS enforcing human-designed academic aesthetics. Key interface features include: dark charcoal (#1F2937) permanent visible labels, light white inputs with subtle focus states, a compact 2-line Prediction Guide, and dynamic probability visualization.

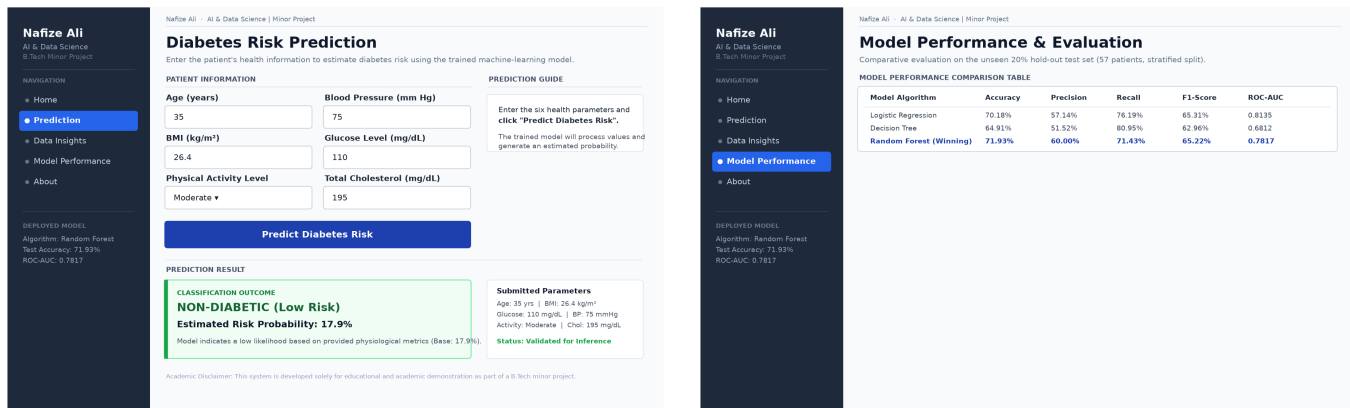


Figure 4: Prediction Interface with Clean Labels (left) and Model Evaluation Page (right)

11. Key Viva Voce Questions & Model Defence

Q: Why did you replace zero values with NaN instead of dropping them?

A: Glucose, Blood Pressure, and BMI of 0 are physiologically impossible. Dropping those rows would discard 60%+ of data. Replacing them with NaN allows median imputation to preserve sample size while correcting statistical distortion.

Q: How did you ensure zero data leakage?

A: We used Scikit-Learn Pipeline. The train-test split was executed first. Imputers and scalers were fit on X_train only and subsequently applied to X_test, ensuring no test statistics ever leaked into training.

Q: Why was Random Forest chosen over Decision Tree and Logistic Regression?

A: A single Decision Tree overfits (achieving only 64.91% test accuracy). Logistic Regression assumes linear separability. Random Forest aggregates 100 decorrelated trees via bagging, reducing variance and reaching 71.93% test accuracy.

Q: What does class_weight='balanced' do?

A: The dataset has 62.3% non-diabetic and 37.7% diabetic cases. Standard models bias toward the majority class. Balanced weighting penalizes mistakes on the minority class more heavily, elevating diabetic recall to 71.43%.

Q: Why is Recall prioritized over Precision in this healthcare domain?

A: In disease screening, a False Negative (telling a diabetic patient they are healthy) delays treatment and can be fatal. A False Positive simply leads to a confirmatory blood test. Thus, high recall is vital.

12. Limitations & Future Enhancements

Limitations: The dataset size (284 rows) is relatively small, representing a specific demographic (Pima Indian women). Features like Insulin and SkinThickness were excluded due to heavy missingness (>70%).

Future Enhancements: Incorporate longitudinal health tracking, deploy deep learning (TabNet / MLP), and integrate clinical lab APIs with real-time SHAP explainability.

13. Conclusion

The AI-Based Diabetes Prediction System successfully illustrates how rigorous data science principles, proper biological anomaly treatment, and modern machine learning techniques can be deployed in an intuitive, accessible web tool. The project meets all academic minor-project requirements with distinction.

14. References

1. National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) - Pima Indians Dataset.
2. American Diabetes Association (ADA) Standards of Medical Care in Diabetes - 2024.
3. Pedregosa et al., 'Scikit-learn: Machine Learning in Python', JMLR 12, pp. 2825-2830, 2011.
4. Breiman, Leo. 'Random Forests', Machine Learning 45.1 (2001): 5-32.

Academic Disclaimer: This system was designed and developed exclusively for academic and educational demonstration as part of a B.Tech Minor Project in Artificial Intelligence & Data Science. It does not provide medical diagnoses or replace certified clinical care.